SAT Hard Question 14

1 ★ Mark for Review **ABC**

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$ respectively. What is the value of x?

2 ★ Mark for Review **ABC**

If a and b are numbers greater than 1 and $\sqrt[7]{a^6}$ is equivalent to $\sqrt{b^3}$, for what value of n is $a^{2n+1} = b$?

3 ★ Mark for Review **ABC**

The function c gives the partial length, in centimeters (cm), of the circumference of a circle with radius r cm:

$$c(r) = \frac{2}{5}(2\pi r)$$

For each increase of 18 cm in r, this partial length increases by $k\pi$ cm. What is the value of k?

4 ★ Mark for Review **ABC**

For each real number r, which of the following points lies on the graph of both equations in the xy-plane for the given system?

$$15x + 45y = 33$$
$$5x + 15y = 11$$

A. $\left(\frac{11r}{5} - 3, r\right)$

B. $\left(-3r + \frac{11}{5}, r\right)$

C. $\left(r, -\frac{r}{3} - \frac{11}{15}\right)$

D. $\left(r, \frac{r}{3} + \frac{11}{15}\right)$

5 ★ Mark for Review ABC

If $\frac{a}{b} = \frac{4}{5}$ and $\frac{5b}{ta} = 35$, what is the value of t?

6 ★ Mark for Review ABC

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c?

$$\frac{1}{2cx} = \frac{x}{16} + \frac{1}{c}$$

7 ★ Mark for Review ABC

The given equation is one equation in a system of two linear equations. If the system has at least one solution, which of the following equations could be the other equation in the system?

$$7x + 5y = 2$$

I. $10.5x + 7.5y = 3$
II. $10.5x - 7.5y = 3$

A. I only

B. II only

C. I and II

D. Neither I nor II

8 ★ Mark for Review ABC

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant. Which of the following could be the value of b?

9 ★ Mark for Review ABC

A right rectangular prism has a base area of $16t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the prism is 9 cm. Which expression represents the surface area, in cm^2, of the right rectangular prism?

10 ★ Mark for Review ABC

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is p% of the mass of object B, what is the value of p?

11 ★ Mark for Review ABC

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

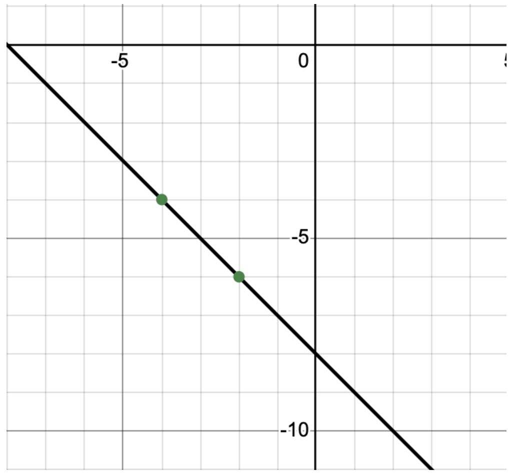
A. $3x^2 + 7x + 14b$

B. $3x^2 + 16x + 14b$

C. $3x^2 + 18x + 14b$

D. $3x^2 + 25x + 14b$

12 ★ Mark for Review ABC



The graph of the linear function $y = f(x) - 19$ is shown. If c and d are positive constants, which equation could define f?

A. $f(x) = d - cx$

B. $f(x) = -d + cx$

C. $f(x) = d + cx$

D. $f(x) = -d - cx$

13 ★ Mark for Review ABC

The expression kx represents the result of decreasing a positive force, x, acting on an object by 0.34%. What is the value of k?

A. 0.34%

B. 1.34%

C. 99.66%

D. 100.34%

14 ★ Mark for Review ABC

A regular polygon has exactly 87 sides. If the measure of each of the 87 interior angles of this polygon is $(180p)^\circ$, what is the value of p?

A. $\frac{85}{87}$

B. $\frac{37}{87}$

C. $\frac{87}{90}$

D. $\frac{87}{180}$

15 ★ Mark for Review ABC

The function $f(n) = 7(20.44)^{\frac{n}{4}}$ gives each term of a sequence as a function of the term's position n in the sequence, where n is a whole number. The value of the term in position 14 is $p\%$ more than the value of the term in position 10. What is the value of p?

16 ★ Mark for Review ABC

A right square pyramid has a total surface area of 28,160 square inches, and the combined surface area of the four lateral faces of this pyramid is 16,060 square inches. What is the height, in inches, of this pyramid?

A. 48

B. 55

C. 73

D. 110

17 ★ Mark for Review **ABC**

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

$$\sqrt{k-x} = 32 - x$$